Sahaj Bhadja

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Summary

Software engineer with hands-on experience in **C/C++ development** for networking and embedded systems. Background in **IP networking fundamentals**, routing protocols, and **Linux systems programming**. Strong foundation in OS internals, data structures, algorithms, and distributed systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Operating Systems, Computer Networks, Analysis of Algorithms, Distributed Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Computer Networks, Operating Systems, Data Structures, Embedded Systems

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Developed high-performance **C++ applications** for data processing pipelines, implementing efficient **memory management** and multi-threaded operations handling 5K+ concurrent requests.
- Optimized system performance through **algorithm analysis** and data structure selection, reducing latency by 35% using lock-free queues and efficient scheduling techniques.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built backend services using **C++ and Python**, implementing efficient data structures for real-time processing with focus on **performance analysis** and optimization.
- Containerized applications using **Docker**, implementing CI/CD pipelines and working with **Linux** environments for development and deployment.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Developed **C++ modules** for high-performance data processing in research pipelines, optimizing memory usage and implementing efficient **scheduling algorithms**.
- Built **distributed systems** components for parallel processing, utilizing multi-threading and inter-process communication for 18% performance improvement.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed **C++ network modules** for real-time data synchronization across distributed nodes, implementing socket programming and **protocol handling** for reliable communication.
- Built **Layer 2/Layer 3 packet processing** components, implementing efficient routing table lookups and forwarding logic that improved throughput by 68%.

NetCore Systems **India**
Embedded Systems Engineer Intern *January 2023 – April 2023*
- Developed **C/C++ software for carrier-grade networking hardware**, implementing **control plane routing table** operations and protocol stack components.
- Worked on **Linux kernel-level** optimizations including scheduling, memory management, and semaphore implementations for embedded networking products.

Skills

Languages: C, C++, Python, SQL, Bash scripting
Networking: IP fundamentals, routing protocols (OSPF, BGP basics), Layer 2/3, socket programming
Systems: Linux kernel, memory management, scheduling, semaphores, multi-threading
Tools: Docker, Git, GDB, Valgrind, Wireshark, CMake, gNMI/NETCONF basics

Projects

RouterSim | *C++, Linux, Socket Programming, Routing Protocols*
- Built a **network router simulator in C++** implementing routing table management, packet forwarding, and basic OSPF/BGP protocol behavior with configurable network topologies.

PacketForge | *C++, Linux, libpcap, Network Protocols*
- Developed a **Layer 2/Layer 3 packet analyzer** using C++ and libpcap, implementing protocol parsing, traffic analysis, and performance metrics for network debugging.

MemScheduler | *C, Linux Kernel, Memory Management, Scheduling*
- Created a **Linux kernel module** implementing custom memory allocation and process scheduling algorithms, demonstrating OS fundamentals including semaphores and context switching.